Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2024-25

Class:	SE	Semester:	III
Course Code:	CSL304	Course Name:	OBJECT-ORIENTED PROGRAMMING IN JAVA

Name of Student:	SHRUTI GAUCHANDRA
Roll No. :	15
Assignment No.:	03
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	3
Legibility	2	1
Total	10	09

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge Legibility	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : MS. NEHA RAUT

Signature :

Date :

Q1. Implement a sequential search algorithm to find a specific element in an array. If the element is found, display its index otherwise indicate that the element is not present in the array.

→ Code:

```
import java.io.*;
import java.util.*;

public class SequentialSearch
{
    public static int sequentialSearch (int[] array,
                                        int target)
    {
        for(int i=0; i<array.length; i++)
        {
            if (array[i] == target)
            {
                return i;
            }
        }
        return -1;
    }

public static void main(String args[])
{
    int[] numbers ={5,3,8,6,2,9,1};
    int target = 6;
    int result = sequentialSearch(numbers, target);
```

```
if (result ! = -1)
{
system.out.println("Element found at index"
                              +result);
}
else
{
System.out.println("Element not present
in                             in the array");
}
}
}
```

Output :

Element found at index: 3.

FOR EDUCATIONAL USE

Q2. Develop a Java program that determines number of days in a given semester. Input to the program year, month & day information of the first and the last days of the semester. The program should accept dates information in a single string instead of accepting year, month & day information seperately. the input string must be in MM/DD/YYYY format. (Assuming semester starts and ends in a year).

→

code:

```
import java.io.*;
import java.time.LocalDate;
import java.time.format.DateTimeFormatter;
import java.time.temporal.ChronoUnit;
import java.util.Scanner;

public class SemesterDaysCalculator
{
  public static void main(String args[])
  {
     Scanner sc = new Scanner(System.in);
   DateTimeFormatter formatter = ~~DateTimeFormatter~~
            DateTimeFormatter.ofPattern("MM/DD/
                                         YYYY");
   System.out.println("Enter the start date of
            the semester (MM/DD/YYYY):");

   String startDateInput = sc.nextLine();
```

```
System.out.println("Enter the end date of the
                    semester (MM/DD/YYYY): ");

String endDateInput = sc.nextLine();

LocalDate startDate = LocalDate.parse(startData
                       input,
                       formatter);

LocalDate endDate = LocalDate.parse
                    (endDateInput, formatter)

long daysBetween =
    ChronoUnit.DAYS.between(startDate,
                    endDate) +1;

System.out.println("The number of days in
       the semester is :" + daysBetween);
}
}
```

Output:

Enter the start date semester : 01/15/2024

Enter the end date semester : 05/10/2024

The number of days in the semester is : 116